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Simulate Static Routing configuration using CISCO Packet Tracer

AIM:

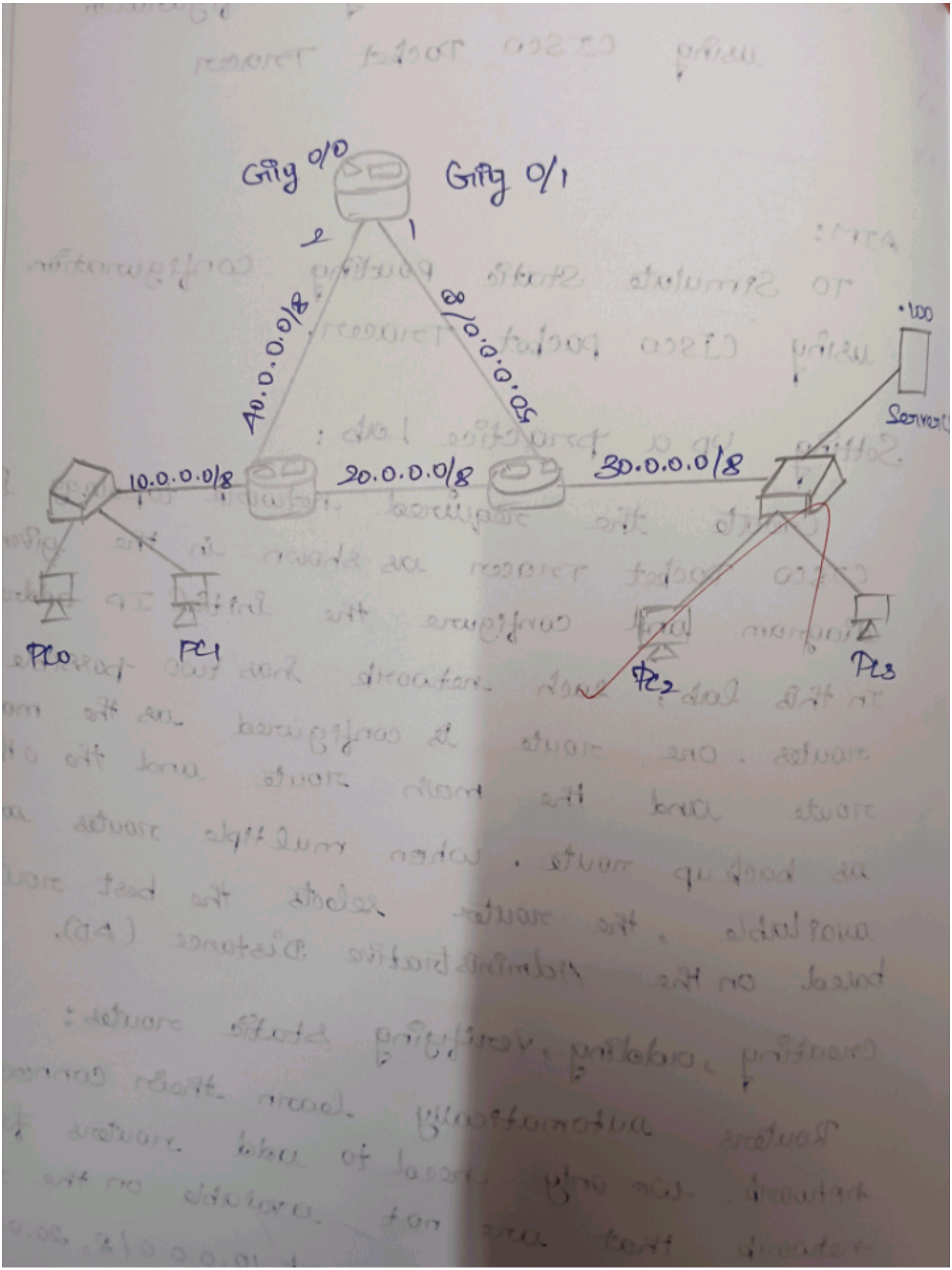
TO Simulate Static Routing configuration using CISCO packet Tracer.

Setting Up a practice Lab:

Create the required network topology in CISCO Packet Tracer as shown in the given diagram and configure the Initial IP address. In this lab, each network has two possible routes. One route is configured as the main route and the main route and the other as backup route. When multiple routes are available, the router selects the best route based on the Administrative Distance (AD).

Creating, adding, Verifying static routes:

Routers automatically learn their connected network. We only need to add routers for the network that are not available on the router interface. For example, network 10.0.0.0/8, 20.0.0.0/8, and 40.0.0.0/8 are directly connected to Routers), Thus, we don't need to configure routes for these network.


Gig 0/0
Gig 0/1
40.0.0.0/8
80.0.0.0/8
.100
Server0
10.0.0.0/8
20.0.0.0/8
30.0.0.0/8
PC0
PC1
PC2
PC3

Router	Available networks on local Interfaces	Networks available on other routers interfaces
Router()	10.0.0.0/8, 20.0.0.0/8, 40.0.0.0/8	30.0.0.0/8, 50.0.0.0/8
Router1	20.0.0.0/8, 30.0.0.0/8, 50.0.0.0/8	10.0.0.0/8, 40.0.0.0/8
Router2	40.0.0.0/8, 50.0.0.0/8	10.0.0.0/8, 20.0.0.0/8, 30.0.0.0/8

Router() Requirements:

* configure two routes for network 30.0.0.0/8, Router1 as the main route and Router2 as the backup route.
* configure two routes for host 30.0.0.100, Router2 as the main route and Router1 as the backup route.
* configure two routes for network 50.0.0.0/8, Router2 as the main route and Router1 as the backup route.
* Verify that only the main routes are added to the routing table.

Router 1 configuration:

Router > enable
Router # configure terminal
Router (config) # ip route 30.0.0.0 255.0.0.0 20.0.0.2 10
Router (Config) # ip route 30.0.0.0 255.0.0.0 40.0.0.2 10

Router (Config) # ip route 30.0.0.0 255.0.0.0 40.0.0.2 10
Router (Config) # ip route 50.0.0.0 255.0.0.0 20.0.0.2 20

Rater (Config) # exist
Router # Show ip route static

Router 2 Requirement:

* Configure a static route for network 10.0.0.0/8 through Router 1.
* Configure a static route for network 30.0.0.0/8 through Router 1.
* Verify that both routes are added to the routing table.

Router 2 configuration:

Router > enable
Router # configure terminal
Router (config) # ip route 10.0.0.0 255.0.0.0 40.0.0.1
Router (Config) # ip route 30.0.0.0 255.0.0.0 50.0.0.2
Router (Config) # exit
Router # Show ip route static

Result:

Static routing was successfully configured and verified in CISCO packet Tracer. The main and backup routes were configured using different AD values, and connectivity between

Bill 14/10